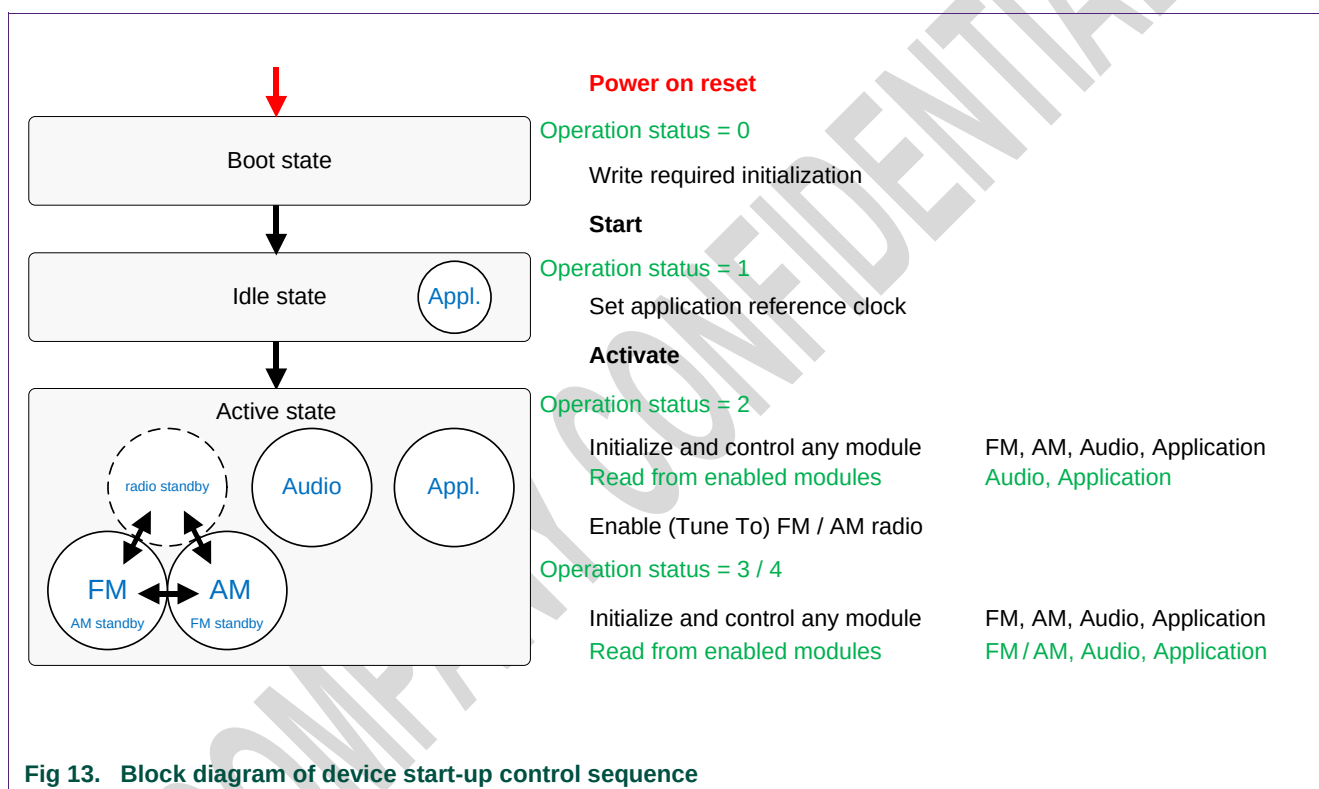


6. Device start-up

6.1 Introduction

The TEF668X is powered by a single supply voltage. No supply power-up requirements are known to exist.

After power-on the device is found in a pre-defined reset state (boot state). A sequence of control transmissions will bring the device into idle state followed by active state to realize the desired user operation mode, function and performance.



Boot state and idle state have no function other than offering the minimum set of controls required for proper active state operation.

State transitions take a small amount of time and care should be taken to ensure a certain state is entered before any of the belonging control transmissions is started. After transmission of a 'Start' or 'Activate' state transition command it is advised to check the operation status by I²C read until the new state is found.

TEF668X state transition times will fall within the following limits:

- Power-on → Boot state : power supply voltage settling + 5 ms.
(or clock active + 5 ms in case of external clock application)
- Boot state → Idle state : 50 ms.
- Idle state → Active state : 100 ms.

6.2 Start-up I²C control transmission sequence

- Supply power on
- Wait until device is found present on the I²C bus ('boot state') :
 - Repeat **APPL_Get_Operation_Status** read until I²C acknowledge from device; status = 0 ('boot state') is found.
`[ w 40 80 01 [ r 0000 ]`
Alternatively Wait radio supply power settling time + 5 ms.
In case of external clock application wait for clock active + 5 ms.
- Send required initializations :
 - See [6.2.1 Required initialization I2C transmission – I2C example](#) and [6.2.2 Required initialization I2C transmission – C code include file](#)
- Start :
 - `[ w 14 0001 ]`
- Wait until device is found in 'idle state' :
 - Repeat **APPL_Get_Operation_Status** read until status = 1 ('idle state') is found.
`[ w 40 80 01 [ r 0001 ]`
Alternatively Wait 50 ms.
- Set reference frequency
 - **APPL_Set_ReferenceClock** freq. = 55466670/9216000/4000000, type = 0/1.
`[ w 40 04 01 034E 5AAE 0001 ]` : 55.46667 MHz external reference.
`[ w 40 04 01 003D 0900 0000 ]` : 4 MHz crystal reference.
Not required for default 9.216 MHz crystal use.
- Activate :
 - **APPL_Activate** mode = 1.
`[ w 40 05 01 0001 ]`
- Wait until device is found in 'active state radio standby' :
 - Repeat **APPL_Get_Operation_Status** read until status = 2 ('radio standby state') is found.
`[ w 40 80 01 [ r 0002 ]`
Alternatively Wait 100 ms.
- Full API control is available from here for all modules.
 Audio and Application are on, FM and AM radio are in low-power standby.
 - Example **FM_**, **AM_**, **AUDIO_** and **APLL_** desired user settings: *module initialization.*
 - Example **AUDIO_Set_Mute** mode = 0 : *disable mute.*
`[ w 30 0B 01 0000 ]`
 - Example **FM_Tune_To** mode = 1, frequency = 8930 : *FM radio 89.3 MHz.*
`[ w 20 01 01 0001 22E2 ]`

6.2.1 Required initialization I²C transmission – I²C example

The following I²C transmissions (version p2.22) are a required part of the V102 boot state device initialization.

Use of these I²C transmissions is required for proper and full function and performance as described in this user manual. Version p2.22 is a minor update over p2.21.

Note: The transmission example below shows a data content of 12 words for every '1B' data transmission. The data stream can be split in any desired length on (2-byte) word boundaries with every data transmission starting with hex value 1B.

```

▪ [ w 1C 0000 ]
  [ w 1C 0074 ]

▪ [ w 1B F000 3838 D080 F000 3840 D080 43B2 3843 D080 F000 7000 C2F7 ]
  [ w 1B F000 3874 D080 80FC 3917 D080 C238 2011 40B7 9FA7 391C D280 ]
  [ w 1B 9FA7 3926 D280 F000 392D D080 F000 3930 D080 9041 3952 DD80 ]
  [ w 1B F000 3A29 D080 F000 3956 D080 F000 3A2B D080 C4A2 0216 6004 ]
  [ w 1B F000 39C3 D080 F000 39EB D080 9001 3958 D080 F000 38F9 D080 ]
  [ w 1B F000 395F D280 F000 3963 D080 4384 396D D080 F000 3970 D080 ]
  [ w 1B F000 3974 D080 9E30 18F9 D280 F000 397A D080 F000 397D D080 ]
  [ w 1B 3077 39E2 D080 F000 3980 D080 3200 3982 D080 F000 3984 D080 ]
  [ w 1B F000 39CB D080 F000 39F9 D080 A801 3A17 D080 F000 3A36 D080 ]
  [ w 1B F000 3A3D D080 F000 3A3F D080 F000 3A53 D080 F000 3A56 D080 ]
  [ w 1B F000 3A59 D080 F000 3A5C D080 F000 7000 F000 3A61 D080 ]
  [ w 1B F000 3A22 D280 F000 3A6F D080 0043 3A7E D980 F000 3A88 D080 ]
  [ w 1B F000 3ADE D080 F000 3AE1 D080 F000 3AF2 D080 2E40 3B90 D080 ]
  [ w 1B F000 3B9A D080 F000 0E3F 6000 5010 28D8 D280 9101 0136 6000 ]
  [ w 1B F000 7000 A018 F000 7000 A0D8 F000 7000 A0F9 F000 7000 A1B5 ]
  [ w 1B F000 7000 A209 F000 7000 A234 F000 7000 A24C F000 2031 D080 ]
  [ w 1B F000 04C1 6008 F000 0101 D280 F000 0030 D080 007F 6002 E200 ]
  [ w 1B F000 0E22 600A F000 00FF 6003 F000 0142 D280 9003 4002 F000 ]
  [ w 1B 9043 0170 D180 F000 0169 D080 0E69 600A A352 2023 0001 6001 ]
  [ w 1B F000 7000 F000 C4CB 7000 F000 CA09 3023 F000 C2CB 7000 F000 ]
  [ w 1B F000 3023 D080 8200 0D50 6008 F000 0D51 6009 3000 2180 6001 ]
  [ w 1B F000 4032 F000 3011 45F3 F000 3092 2D30 6004 3113 2D40 6005 ]
  [ w 1B 3194 7FFF 6006 3215 0D61 600A 3296 0D6B 600B 3310 0D50 6001 ]
  [ w 1B 3390 0D5C 6002 3021 0D63 6003 3031 0D75 600C 30A2 8D00 6001 ]
  [ w 1B 30B3 0173 6002 3041 0025 6003 30C2 4044 F000 3143 4035 F000 ]
  [ w 1B 31C4 6400 6006 3245 1F40 6007 32C6 7000 F000 3347 1EBC 600D ]
  [ w 1B 33C0 0122 6001 3440 FDEE 6002 3051 7B8F 6003 30D2 C429 6004 ]
  [ w 1B 3151 1EC2 600E 3253 FF0D 6002 32D4 7D2E 6003 3061 C19A 6004 ]
  [ w 1B 30E2 7000 F000 3161 7000 F000 3263 7000 F000 32E4 7000 D008 ]
  [ w 1B F000 0370 D280 F000 7000 A002 F000 7000 A059 F000 0215 D080 ]
  [ w 1B F000 0D51 600F F000 0517 600E 23F6 7000 F000 F000 2163 41F5 ]
  [ w 1B 918F 21F8 4074 C3EF 21E0 F000 C3A4 33F7 F000 D85B 7000 F000 ]
  [ w 1B 8218 7000 F000 9FAF 1800 F000 9F0F 31F8 9002 F000 7000 9028 ]
  [ w 1B F000 7000 D008 F000 2278 F000 16D3 6009 A07F 35F0 1EBC 600D ]
  [ w 1B F000 0D61 6008 F000 03A5 D280 F000 1EC2 600D F000 0D6B 6008 ]
  [ w 1B F000 03A5 D280 F000 2100 F000 836D 22F1 F000 F000 2377 F000 ]
  [ w 1B 9041 3670 F000 9E79 7000 9001 F000 32F1 D008 91C7 3375 F000 ]
  [ w 1B F000 3470 E600 F000 34F0 E600 F000 2474 F000 F000 24F3 F000 ]
  [ w 1B 8C24 26F2 4016 8A1B 3474 4FF5 82B7 34F3 F000 F000 2071 9005 ]
  [ w 1B 8304 7000 F000 8E67 7000 F000 F000 7000 9002 F000 36F6 F000 ]
  [ w 1B F000 34F0 8006 82AF 7000 F000 821B 7000 D009 8E5F 7000 F000 ]
  [ w 1B F000 7000 D009 F000 36F5 F000 F000 3470 F000 4011 2772 A2F5 ]
  [ w 1B 908A 20F3 A2F4 8ED7 3772 F000 F000 37F1 E600 F000 7000 D008 ]
  [ w 1B F000 227A F000 16C3 6009 A058 F000 1820 F000 F000 3570 F000 ]
  [ w 1B F000 327A D008 8200 0D51 6008 4003 7000 F000 3380 7000 F000 ]
  [ w 1B 2106 7000 F000 3700 7000 F000 3780 4015 F000 3683 7000 F000 ]
  [ w 1B 3305 0D61 6009 3286 0D6B 600A 3210 7000 F000 3290 7000 F000 ]
  [ w 1B 3310 7000 F000 3390 7000 F000 3410 7000 F000 3490 7000 F000 ]
  [ w 1B 3110 7000 F000 3190 7000 F000 3220 7000 F000 32A0 7000 F000 ]
  [ w 1B 3320 7000 F000 33A0 7000 F000 3420 7000 F000 34A0 7000 F000 ]
  [ w 1B 3120 7000 F000 31A0 7000 F000 8200 0D30 600A 0D40 600B C010 ]
  [ w 1B F000 1020 F000 0D51 600C C010 F000 1030 F000 F000 35C0 D008 ]
  [ w 1B F000 0D75 600F F000 0563 600E 24F7 051D 600D 2576 7000 F000 ]
  [ w 1B 91C7 20E8 4015 918F 21E9 D409 C3EF 2000 4012 9FBE 2011 5803 ]
  [ w 1B A080 3577 9001 F000 7000 D008 F000 21F5 F000 A0CA 2254 F000 ]

```

[w 1B CC09 0517 600C 832C 7000 F000 8A61 7000 F000 AE48 2245 A2BD]
[w 1B A228 2078 F000 F000 35F0 F000 F000 1800 F000 F000 3078 F000]
[w 1B 16E3 6009 A027 8901 23F4 F000 F000 20F2 F000 8261 2173 F000]
[w 1B A050 3670 F000 A058 2372 E140 A801 22F3 F000 9049 2275 E040]
[w 1B 8061 7000 F000 8A51 33F1 F000 A058 7000 F000 AF48 7000 F000]
[w 1B F000 3470 D008 8200 0D75 6008 9009 0D00 6009 F000 3500 F000]
[w 1B F000 3380 C028 F000 1010 F000 F000 3481 D008 8249 0D75 6008]
[w 1B F000 7000 8FFD 0400 6000 A2A3 8EC0 4000 6005 6000 6005 E600]
[w 1B C81B 7000 F000 D8DB 0D51 6008 835B 7000 F000 9EBA 3003 F000]
[w 1B F000 3084 D409 F000 7000 8FAF F000 0D75 6008 F000 0D51 6009]
[w 1B F000 2403 F000 F000 2794 D008 A003 7000 F000 0011 0800 F000]
[w 1B 0011 0800 C00E A009 0011 0800 A009 7000 F000 A408 7000 D008]
[w 1B A003 7000 F000 0011 0800 F000 0011 0800 C026 A009 0011 0800]
[w 1B A009 7000 F000 A408 7000 D008 F000 1D01 6008 F000 0A2C 6000]
[w 1B F000 011A 6001 3100 7000 F000 3181 7000 D008 A801 7FFF 6006]
[w 1B CC0A 7000 F000 8EA1 3106 F000 F000 3206 D409 AEE8 04DF D080]
[w 1B 9087 7000 9807 F000 0FB1 D280 9E08 4189 6006 1492 6000 9004]
[w 1B A198 1D01 6008 AA56 7000 F000 A220 3206 F000 F000 3280 F000]
[w 1B 91C2 069A 6008 41E6 0600 D080 8213 7000 D809 F000 0EB1 D280]
[w 1B 9E08 1062 6001 051F 6005 D009 F000 06E6 6008 A365 7000 F000]
[w 1B F000 3085 D008 1000 6003 A26F 3023 0773 D280 F000 07C6 D080]
[w 1B 0800 6003 A002 0E6F 6009 F000 0200 6003 8000 F000 0773 D080]
[w 1B 40E0 001F 6001 13D5 6007 A011 9040 7000 F000 13FB 6006 A00E]
[w 1B 1405 6006 A00D 140F 6006 A00C 4560 003A 6001 13B6 6007 A00B]
[w 1B 9040 7000 F000 13F7 6006 A008 1401 6006 A007 140B 6006 A006]
[w 1B 4180 003B 6001 13B6 6007 A005 9040 7000 F000 13F7 6006 A002]
[w 1B 1401 6006 A001 140B 6006 8000 F000 0D28 D080 D7CA 00FF 6004]
[w 1B 81D7 0CF7 6009 D056 7000 F000 8276 3017 F000 D0F6 4083 F000]
[w 1B C1A4 2019 F000 82F6 7000 F000 C180 2017 A24E C3E7 7000 F000]
[w 1B C5C7 7000 F000 F000 3017 D008 9A78 7000 F000 9A70 7000 90DE]
[w 1B F000 7000 90DE F000 0D67 D080 F000 7000 A114 F000 7000 80CD]
[w 1B 1EC8 6008 A244 F000 2000 A243 9000 7000 F000 F000 7000 D409]
[w 1B F000 7000 8FA5 1EC8 6008 A23F F000 3000 D008 0028 6000 A23D]
[w 1B 9E38 0EF4 6009 9E38 7000 9FFF F000 7000 D008 4040 0C8A D580]
[w 1B F000 0C8B D280 F000 0FB1 D280 9C39 7000 F000 9C31 7000 9003]
[w 1B F000 7000 9002 4010 7000 AFF3 41F1 4040 800D 4000 7000 AFF1]
[w 1B 41F1 4040 800C 03E8 6002 A22F 9083 1044 D180 F000 1042 D080]
[w 1B 4071 4020 A008 43A1 4030 A007 43B1 4030 A006 F000 1075 D080]
[w 1B 4071 4020 A004 43A1 4030 A002 43B1 4030 A001 F000 1081 D080]
[w 1B F000 0C92 D080 F000 0C94 D080 F000 0BC9 6008 F000 1D8D D280]
[w 1B F000 16D5 D080 F000 0BC9 6008 F000 1D8F D280 F000 16DA D080]
[w 1B 3E91 1F5F D280 F000 238A D080 0D85 6008 A21A F000 3000 D008]
[w 1B F000 0F88 600B F000 0230 6003 F000 0938 6004 9F98 012C 6005]
[w 1B F000 2EB3 EE00 F000 0FB1 D280 9E09 0D87 600C 9F90 7000 9002]
[w 1B 9F88 2F34 EE00 F000 2FB5 EE00 3043 0B42 6008 F000 30C4 F000]
[w 1B F000 3145 F000 F000 2CB1 4017 F000 2D32 F000 C679 2205 F000]
[w 1B F000 3D81 F000 F000 3D01 F000 DD6D 1B93 D280 9046 0D85 600A]
[w 1B 2042 1B93 D280 DAC9 2025 A203 8C69 2DB3 F000 DD49 3201 F000]
[w 1B 8D89 7000 F000 3C81 FE35 6000 3581 0016 6001 A018 2146 F000]
[w 1B AA07 012D 6000 81CF 7000 F000 8F80 7000 F000 F000 7000 9402]
[w 1B 4045 574B 6000 E000 6001 8002 4065 4000 6000 C666 6001 8000]
[w 1B A030 4000 6004 AF4E 00A4 6005 818E 0B75 6008 F000 1474 D280]
[w 1B F000 1B90 D280 2E34 0400 6006 20C2 1B93 D280 D2C9 03E9 6005]
[w 1B 818E 0B59 6008 8F2D 31C6 F000 F000 7000 9402 4045 B15C 6003]
[w 1B 4666 6007 8002 4065 AACE 6003 C666 6007 8000 A0E0 0038 6004]
[w 1B AF48 001D 6005 81C7 3184 F000 3105 1474 D280 F000 1B90 D280]
[w 1B F000 D800 6000 F000 0B67 6008 21C6 2000 6004 F000 1333 6005]
[w 1B 8186 3184 4007 3105 1474 D280 F000 1B90 D080 F000 6E6C 6003]
[w 1B 4007 00E5 6005 A058 FFFF 6003 AA24 0B52 6008 811C 7000 F000]
[w 1B 3105 7FDF 6006 3184 1474 D280 F000 1B90 D080 9183 2E85 F000]
[w 1B F000 2F06 F000 F000 2F87 F000 0BC9 6009 A1CE 3E11 0D82 600A]
[w 1B F000 2711 A1CC A050 2792 F000 AA41 0BB4 600C 8052 2021 A1C9]
[w 1B 9ECB 3592 F000 9041 7000 D809 882D 7000 D409 F000 3615 F000]
[w 1B F000 3695 F000 9100 30A6 F000 3127 1462 D280 9004 31C0 F000]
[w 1B 2145 8CCD 6006 F000 FF55 6007 F000 1474 D280 F000 3240 F000]
[w 1B F000 32C1 F000 F000 3342 8000 F000 0D82 6009 F000 0BC9 600A]
[w 1B F000 2015 A1B8 9145 7FEC 6006 4047 7000 D409 2090 1447 D280]
[w 1B F000 34A0 F000 2110 1447 D280 F000 3520 D008 0A3D 6007 A1B1]
[w 1B A279 0BC9 6009 A2BA 7000 F000 90C0 3B91 F000 3B12 1462 D280]
[w 1B F000 3C10 D008 F000 0B60 600E F000 0004 6003 40A7 3FFC 6004]
[w 1B 3263 8008 6005 32E4 199A 6006 3365 0D86 600F 31E6 7000 F000]

[w 1B 3077 7000 D008 0D82 600A A1A3 F000 3020 F000 00D2 6001 940B]
[w 1B F000 00D2 6002 00FA 6003 AFED F000 BE77 6006 F000 4B00 6007]
[w 1B 3706 0001 6001 3787 03E8 6002 F000 0384 6004 F000 0001 6005]
[w 1B F000 008C 6006 F000 0AF0 6007 F000 7000 8FC7 F000 0BB4 6009]
[w 1B F000 0011 6005 F000 0011 6006 3B05 1000 6007 3B86 0004 6000]
[w 1B 3C07 AD84 6001 3210 01FD 6002 3291 0B5E 6003 3312 0C80 6004]
[w 1B 3193 00A0 6005 3484 FFD0 6006 3505 E354 6007 3606 3400 6000]
[w 1B 3686 0001 6001 3707 03E8 6002 3780 7000 8FB7 80E5 7FFF 6000]
[w 1B CC0A 7000 F000 8EA9 3370 F000 AF20 3470 D409 AE88 7000 F000]
[w 1B A805 0080 6007 A8F6 3375 F000 C386 7000 F000 8F7F 33F6 F000]
[w 1B A885 1DE4 D180 F000 3475 D008 0D86 6009 A17A 2012 1F32 D280]
[w 1B F000 3582 D008 9082 0D86 6009 40A7 4077 E640 F000 7000 F000]
[w 1B F000 3017 D008 0BB4 6008 A173 F000 3182 D008 F000 0961 6008]
[w 1B F000 09C9 6009 F000 2180 F000 F000 2091 A16E A208 7000 D008]
[w 1B 00CF 6002 A001 0151 6002 8001 0D8B 6008 8001 0D8C 6008 8000]
[w 1B F000 7000 F000 F000 3002 D008 0FF2 6009 A166 2011 0D8B 600A]
[w 1B F000 2C92 F000 9041 2011 F000 F000 7000 93FE F000 7000 F200]
[w 1B 3122 2031 D080 0D8B 6008 A003 F000 2022 D080 0D8C 6008 A001]
[w 1B F000 201E D080 F000 7000 F000 2000 0E69 6008 F000 01D0 6001]
[w 1B F000 3080 F000 F000 3001 D008 836D 0C35 6008 4060 3A51 6001]
[w 1B 41E2 2196 6003 3300 4144 F000 3381 7000 F000 3402 7000 F000]
[w 1B 3483 0C29 6009 3504 3A5F 6000 3585 3A64 6003 3090 7000 F000]
[w 1B 3393 7000 8FDF F000 7000 AE5F F000 7000 8EA4 F000 7000 AE9D]
[w 1B F000 0C51 D280 F000 21A0 D080 F000 052E D280 F000 7000 A00E]
[w 1B F000 21B7 D080 F000 1F32 D280 F000 1E84 D280 F000 21E0 D080]
[w 1B F000 1EA4 D280 F000 07F7 D280 F000 2215 D080 F000 07FB D280]
[w 1B F000 228F D080 4040 22E1 D280 4080 22E2 D280 F000 229C D080]
[w 1B F000 07FB D280 F000 22B6 D080 F000 20F0 D280 9002 27DF D280]
[w 1B 9E69 7000 F000 F000 0700 D180 F000 2320 D080 9082 0C29 6009]
[w 1B 4017 4067 E640 F000 7000 F000 F000 3217 D008 F000 07F7 D280]
[w 1B F000 27EA D280 F000 11B5 D080 4055 1517 6008 F000 1D40 6000]
[w 1B 3105 1E60 6001 3185 7000 F000 3205 7000 F000 3285 7000 F000]
[w 1B 3300 7000 F000 3381 0D8E 6009 3400 0254 6002 3481 3880 6003]
[w 1B 3012 7000 F000 3093 7000 D008 170B 600C A11E 0040 4005 F000]
[w 1B 0041 243F D080 F000 0C83 6008 9000 0C6B 600B F000 700C E600]
[w 1B F000 2003 F000 F000 2204 F000 F000 3BB3 F000 F000 3C34 D008]
[w 1B F000 7000 ADC1 F000 0C8D 6008 F000 260A D080 83FF 0D9B 6008]
[w 1B F000 01F4 6000 F000 03B1 6001 1000 03B2 6002 1001 040E 6000]
[w 1B 1002 040F 6001 1000 045C 6002 1001 045D 6000 1002 1380 6001]
[w 1B 1000 0DAB 6009 1001 02EE 6000 1007 4306 6001 1010 0469 6002]
[w 1B 1011 4487 6000 1012 05E3 6001 1010 4608 6002 1011 06AE 6000]
[w 1B 1012 0DA4 6008 1010 9E3C 6000 1017 0DB3 6009 1000 25DA 6000]
[w 1B 1007 803D 6001 1010 2882 6002 1011 C03D 6000 1012 2887 6001]
[w 1B 1010 C03B 6002 1011 288C 6000 1012 C03B 6001 1010 2891 6002]
[w 1B 1011 C03B 6000 1012 2896 6001 1010 C03B 6002 1011 289B 6000]
[w 1B 1012 C03B 6001 1010 28A0 6002 1011 403C 6000 1012 28A5 6001]
[w 1B 1010 C03C 6002 1011 28AA 6000 1012 C03C 6001 1010 28AF 6002]
[w 1B 1011 C03C 6000 1012 28B4 6001 1010 C03C 6002 1011 28B9 6000]
[w 1B 1012 C03C 6001 1010 28BE 6002 1011 C03C 6000 1012 29CC 6001]
[w 1B 1010 0037 6002 1011 29D1 6000 1012 0037 6001 1010 29D6 6002]
[w 1B 1011 0037 6000 1012 29DB 6001 1010 0037 6002 1011 0DD8 6008]
[w 1B 1012 2715 6000 1017 403E 6001 1000 271A 6002 1001 803C 6000]
[w 1B 1002 2864 6001 1000 403C 6002 1001 2869 6000 1002 403C 6001]
[w 1B 1000 286E 6002 1001 803A 6000 1002 2873 6001 1000 803A 6002]
[w 1B 1001 28A0 6000 1002 403C 6001 1000 29CC 6002 1001 0037 6000]
[w 1B 1002 29D1 6001 1000 0037 6002 1001 29D6 6000 1002 0037 6001]
[w 1B 1000 29DB 6002 1001 0037 6000 1002 0DEF 6009 1000 7000 F000]
[w 1B 1007 7000 F000 1017 7000 D008 0CDF 600B A0BE 3030 23D6 D280]
[w 1B F000 26E8 D080 0D9B 6008 A0BB F000 0002 A0BA 9082 7000 F000]
[w 1B 828A 7000 9003 908A 7000 9001 F000 7000 8FFB 82BF 7000 F000]
[w 1B F000 0199 D280 F000 0D9A 6008 F000 26C1 F000 F000 0002 A0B1]
[w 1B 9082 7000 F000 828A 7000 9003 908A 7000 9001 F000 7000 8FFB]
[w 1B 8280 7000 F000 F000 7000 D008 F000 0DAB 600D F000 3FFF 6002]
[w 1B F000 0051 A0A8 C253 7000 F000 8EC4 7000 9001 F000 7000 97FC]
[w 1B D48F 0199 D280 4005 0DA4 600D 4017 3FFF 6002 9F7D 0051 A0A1]
[w 1B C253 7000 F000 82C4 271C D180 F000 7000 97FC 9146 2720 D080]
[w 1B 4007 0199 D280 F000 0D8E 600E 4007 25C1 A09A 9E79 2E40 4014]
[w 1B 4036 3167 9019 2F45 1964 6002 4FF0 1A5E 6003 8354 4036 F000]
[w 1B 835C 4000 E200 F000 4010 E200 F000 7000 F000 9000 7000 F000]
[w 1B F000 7000 9C11 F000 0DD8 600D 4036 22F0 D280 0DB3 600D E600]
[w 1B 4014 2140 F000 F000 2051 A08B 9041 7000 F000 8245 7000 901D]

```

[ w 1B F000 7000 9001 F000 70DC 8FFB 20D1 3FFF 6002 F000 0CDF 600B ]
[ w 1B C253 3161 F000 D48E 3033 F000 D44A 37C6 F000 C6A2 7000 F000 ]
[ w 1B C3A0 3D42 F000 F000 31E0 D008 C624 31E0 F000 9000 3D44 F000 ]
[ w 1B 83B6 7000 E040 40D5 2141 A07B CC0B 4000 F000 80ED 23EA D280 ]
[ w 1B BF60 2060 F000 F000 20E1 F000 F000 23EA D280 90C6 23DE D280 ]
[ w 1B 4F95 4014 A074 8175 7000 F000 AF43 7000 F000 C2E0 7000 F000 ]
[ w 1B 80C3 0CDF 600B F000 3363 F000 F000 34E3 F000 F000 3033 A041 ]
[ w 1B F000 7000 8012 40D5 2141 A06B CC0B 4000 F000 80ED 23EA D280 ]
[ w 1B BF60 2060 F000 F000 20E1 F000 F000 23EA D280 90C6 23DE D280 ]
[ w 1B 4F95 4014 A064 8175 7000 4006 AF43 7000 F000 C2E0 0CDF 600B ]
[ w 1B F000 31E0 F000 80C3 4036 E600 C624 3033 F000 F000 3363 F000 ]
[ w 1B F000 34E3 F000 F000 3D44 F000 F000 37C6 A02E F000 2366 A031 ]
[ w 1B 8292 0BB8 6003 47F4 23E0 D280 5208 6005 9803 8292 5208 6003 ]
[ w 1B 47F4 23E0 D280 5208 6005 9823 F000 33E4 F000 F000 34E6 F000 ]
[ w 1B F000 3564 F000 F000 35E5 F000 F000 2366 A04E 918E 7000 F000 ]
[ w 1B F000 3366 A024 8292 5208 6003 F000 23E0 D280 F000 7000 9C05 ]
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[ w 1B F000 7000 8FF5 F000 24E6 A043 9FBE 7000 F000 F000 34E6 A019 ]
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[ w 1B 4014 23E1 A038 C263 2562 F000 F000 2465 9007 C2A3 25E6 F000 ]
[ w 1B 9080 7000 9002 8377 7000 F000 2560 23E0 EBC0 0CDF 600B A031 ]
[ w 1B F000 3030 F000 F000 21E0 D008 C2A3 25E6 F000 F000 7000 93F9 ]
[ w 1B 9040 7000 8FFA F000 21E0 AF0F F000 2462 D280 F000 3260 F000 ]
[ w 1B F000 32E1 D008 F000 23ED D280 B868 2260 F000 F000 22E1 F000 ]
[ w 1B F000 23EA D280 F000 23DE D280 A988 7000 F000 ABF8 23E4 D280 ]
[ w 1B 4643 2142 A020 A098 7000 F000 F000 23E7 D280 F000 23C6 D280 ]
[ w 1B 9000 7000 F000 F000 7000 DC09 F000 23CE D080 47E5 7000 F000 ]
[ w 1B 4016 20D3 A018 835D 2052 F000 C2F5 2154 9807 C2B6 7000 9002 ]
[ w 1B F000 7000 9005 F000 7000 8002 C2B6 7000 F000 F000 7000 D409 ]
[ w 1B 8264 7000 F000 F000 7000 DC09 F000 3151 F000 F000 30D2 D008 ]
[ w 1B 0DEF 600B A00C 9E79 2036 F000 F000 2045 F000 9FBC 7000 E040 ]
[ w 1B 9145 7000 D009 2644 272D D580 9186 7000 F000 F000 2732 D580 ]
[ w 1B F000 7000 AF67 2F41 2736 D080 F000 2915 D280 F000 3B9E D280 ]
[ w 1B F000 2888 D080 F000 7000 D008 F000 16C3 6008 F000 009A 6000 ]
[ w 1B F000 02E3 6000 1000 04F4 6000 1000 06FF 6000 1000 0907 6000 ]
[ w 1B 1000 0B10 6000 1000 0D1F 6000 1000 0F65 6000 1000 0F65 6000 ]
[ w 1B 1000 0D1F 6000 1000 0B10 6000 1000 0907 6000 1000 06FF 6000 ]
[ w 1B 1000 04F4 6000 1000 02E3 6000 1000 009A 6000 F000 1000 F000 ]
[ w 1B F000 1000 F000 F000 16D3 6008 F000 FF93 6000 F000 FE37 6000 ]
[ w 1B 1000 FDD9 6000 1000 FE9F 6000 1000 0385 6000 1000 0D6B 6000 ]
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[ w 1B 1000 FE37 6000 1000 FF93 6000 F000 1000 F000 1000 F000 ]
[ w 1B F000 16E3 6008 F000 0064 6000 F000 FFA8 6000 1000 FFA6 6000 ]
[ w 1B 1000 FFDF 6000 1000 0001 6000 1000 0128 6000 1000 012F 6000 ]
[ w 1B 1000 FD23 6000 1000 FDA1 6000 1000 0389 6000 1000 0254 6000 ]
[ w 1B 1000 FE7D 6000 1000 008C 6000 1000 FBE1 6000 1000 F942 6000 ]
[ w 1B 1000 0C47 6000 1000 0EA2 6000 1000 EC43 6000 1000 EACE 6000 ]
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[ w 1B 1000 0EA2 6000 1000 0C47 6000 1000 F942 6000 1000 FBE1 6000 ]
[ w 1B 1000 008C 6000 1000 FE7D 6000 1000 0254 6000 1000 0389 6000 ]
[ w 1B 1000 FDA1 6000 1000 FD23 6000 1000 012F 6000 1000 0128 6000 ]
[ w 1B 1000 0001 6000 1000 FFDF 6000 1000 FFA6 6000 1000 FFA8 6000 ]
[ w 1B 1000 0064 6000 F000 1000 F000 F000 1000 F000 F000 170B 6008 ]
[ w 1B F000 0003 6000 F000 54C0 6000 1000 0005 6000 1000 0005 6000 ]
[ w 1B 1000 000F 6000 1000 000F 6000 1000 09C0 6000 1000 0A20 6000 ]
[ w 1B 1000 1D40 6000 1000 1E60 6000 F000 1000 F000 F000 1000 F000 ]
[ w 1B F000 7000 D008 ]

[ w 1C 0000 ]
[ w 1C 0075 ]

[ w 1B 4013 402F 4168 41C1 4214 44DE 44E9 45FB 4611 47C5 47FC 4D5D ]
[ w 1B 4E56 4E58 4E5B 4D83 4E0A 4E0B 4E49 4E53 4F92 4FEA 5041 5074 ]
[ w 1B 5080 5660 56D4 56D9 5829 59B4 5A3A 5B79 5BA2 5BEF 5DDC 603A ]
[ w 1B 6088 60A8 619F 61B6 61DF 6214 6259 629B 62A1 6369 643B 6609 ]
[ w 1B 66E7 670B 671A 6729 6887 6899 68A7 68B2 ]

[ w 1C 0000 ]

```

The following data sets (p2.22) are a required part of boot state device initialization.

The C code in this chapter can be included in C program code of a host μ C and

describes the same data as the I²C script example of chapter 6.2.1.

The C code may be copied from this document but the C code include file can also be

found as part of the PC GUI control program installation (.../documentation/PATCH.c).

```

/*
 * Copyright Catena / NXP semiconductors
 * Creation date: 2015 02 16 9:57:36
 * Version: 2.00 - p2.22
 *
 * Usage:
 * - Compile and link this C-file and declare in the source-file for required initialization transmissions the following:
 *
 * extern const size_t lutsSize;
 * extern const unsigned char * plutBytes;
 *
 * extern const size_t PatchSize;
 * extern const unsigned char * pPatchBytes;
 *
 * The source for required initialization transmissions should create the following set of I2C transmissions:
 *
 * [ w 1C 0000 ] script example: S C8 1C 00 00 P
 * [ w 1C 0074 ] script example: S C8 1C 00 74 P
 * [ w 1B (pPatchBytes) ] script example: S C8 1B F0 00 .... P
 * PatchSize of pPatchBytes can be divided (on a 2 byte boundary) in separate transmissions all starting with command identifier 0x1B.
 * [ w 1C 0000 ] script example: S C8 1C 00 00 P
 * [ w 1C 0075 ] script example: S C8 1C 00 75 P
 * [ w 1B (plutBytes) ] script example: S C8 1B 40 13 .... P
 * lutsSize of plutBytes can be divided (on a 2 byte boundary) in separate transmissions all starting with command identifier 0x1B.
 * [ w 1C 0000 ] script example: S C8 1C 00 00 P
 */

#include <stdlib.h>

extern const size_t PatchSize;
extern const unsigned char * pPatchBytes;

static const unsigned char PatchByteValues[] = {
(
0xF0, 0x00, 0x38, 0x38, 0x00, 0x80, 0xF0, 0x00, 0x38, 0x40, 0x00, 0x80, 0x43, 0x82, 0x38, 0x43, 0x00, 0x80, 0xF0, 0x00, 0x40, 0x00, 0xC2, 0xF7,
0xF0, 0x00, 0x38, 0x74, 0x00, 0x80, 0x80, 0xFC, 0x39, 0x17, 0x00, 0x80, 0xC2, 0x38, 0x20, 0x11, 0x40, 0x87, 0x9F, 0x87, 0x39, 0x01, 0xC2, 0x80,
0x9F, 0xA7, 0x39, 0x26, 0xD2, 0x80, 0xF0, 0x00, 0x39, 0x2D, 0x00, 0x80, 0x00, 0x39, 0x30, 0x00, 0x80, 0x90, 0x41, 0x39, 0x52, 0xD2, 0x80,
0xF0, 0x00, 0x3A, 0x29, 0x00, 0x80, 0xF0, 0x00, 0x39, 0x56, 0x00, 0x80, 0xF0, 0x00, 0x3A, 0x2B, 0x00, 0x80, 0xC4, 0xA2, 0x02, 0x16, 0x60, 0x04,
0xF0, 0x00, 0x39, 0xC3, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80,
0xF0, 0x00, 0x39, 0x5F, 0xD2, 0x80, 0xF0, 0x00, 0x39, 0x63, 0xD2, 0x87, 0x8A, 0x8A, 0x39, 0x6D, 0xC0, 0xF0, 0x00, 0x39, 0x70, 0x00, 0x80,
0xF0, 0x00, 0x39, 0x74, 0x00, 0x80, 0x80, 0x30, 0x18, 0x8F, 0xD2, 0x80, 0xF0, 0x00, 0x39, 0x7A, 0x00, 0x80, 0x00, 0x39, 0x70, 0x00, 0x80, 0x39, 0x70, 0x00, 0x80,
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0xF0, 0x00, 0x39, 0x80, 0x00, 0x80, 0x00, 0x30, 0x00, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80,
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00F0, 0000, 0009, 00C9, 0060, 0009, 00F0, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000,
0000, 000C, 0060, 0002, 00A0, 0001, 0001, 0051, 0060, 0002, 0080, 0001, 0000, 008B, 0060, 0008, 0000, 0001, 0000, 008C, 0060, 0000, 0080, 0000,
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00F0, 0000, 002C, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000,
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00F0, 0000, 0020, 001E, 0000, 0080, 00F0, 0000, 0000, 0070, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000, 0000,
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0xE2, 0x64, 0x70, 0x00, 0xF0, 0x00, 0xF0, 0x00, 0x70, 0x00, 0xDC, 0x09, 0xF0, 0x00, 0x31, 0x51, 0xF0, 0x00, 0xF0, 0x00, 0x30, 0xD2, 0x00, 0xB8,
0xD0, 0xBF, 0x60, 0x08, 0xA0, 0xDC, 0xBE, 0x79, 0x20, 0x36, 0xF0, 0x00, 0xF0, 0x00, 0x20, 0x45, 0xF0, 0x00, 0xFF, 0xBC, 0x70, 0x00, 0xE0, 0x40,
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0xF0, 0x00, 0x02, 0xE3, 0x60, 0xD0, 0x10, 0xD0, 0x04, 0xF4, 0x60, 0xD0, 0x10, 0xD0, 0x06, 0xFF, 0x60, 0xD0, 0x10, 0xD0, 0x09, 0x07, 0x60, 0xD0,
0x10, 0xD0, 0x08, 0x10, 0x60, 0xD0, 0x10, 0xD0, 0xD0, 0x1F, 0x60, 0xD0, 0x10, 0xD0, 0x08, 0x10, 0x60, 0xD0, 0x10, 0xD0, 0x09, 0x07, 0x60, 0xD0, 0x10, 0xD0, 0x06, 0xFF, 0x60, 0xD0,
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0x10, 0xD0, 0xFF, 0x7D, 0x60, 0xD0, 0x10, 0xD0, 0xD0, 0x8C, 0x60, 0xD0, 0x10, 0xD0, 0xF0, 0xB1, 0x60, 0xD0, 0x10, 0xD0, 0xF0, 0x00, 0xF0, 0x00,
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0x10, 0xD0, 0x00, 0x0F, 0x60, 0xD0, 0x10, 0xD0, 0xD0, 0x00, 0x0F, 0x60, 0xD0, 0x10, 0xD0, 0x09, 0xC0, 0x60, 0xD0, 0x10, 0xD0, 0x0A, 0x20, 0x60, 0xD0,
0x10, 0xD0, 0x1D, 0x40, 0x60, 0xD0, 0x10, 0xD0, 0x1E, 0x60, 0xD0, 0xF0, 0x00, 0x10, 0xD0, 0xF0, 0x00, 0xF0, 0x00, 0x10, 0xD0, 0xF0, 0x00,
0xF0, 0x00, 0x70, 0x00, 0xD0, 0x08
```

```
};

const size_t PatchSize = sizeof(PatchByteValues);
const unsigned char * pPatchBytes = &PatchByteValues[0];

extern const size_t LutSize;
extern const unsigned char * pLutBytes;

static unsigned char LutByteValues[] =
{
    0x40, 0x13, 0x40, 0x2F, 0x41, 0x68, 0x41, 0xC1,
    0x42, 0x14, 0x44, 0xD8, 0x44, 0xB9, 0x45, 0xF8,
    0x46, 0x11, 0x47, 0xC5, 0x47, 0xFF, 0x40, 0x50,
    0x48, 0x56, 0x48, 0x58, 0x48, 0x5B, 0x4D, 0x83,
    0x4E, 0x0A, 0x4E, 0x0B, 0x4E, 0x49, 0x4E, 0x53,
    0x4F, 0x52, 0x4F, 0x5A, 0x50, 0x41, 0x50, 0x74,
    0x50, 0x50, 0x56, 0x60, 0x56, 0x04, 0x56, 0x09,
    0x58, 0x29, 0x59, 0x84, 0x5A, 0x3A, 0x5B, 0x79,
    0x5B, 0xA2, 0x5B, 0xBF, 0x5D, 0xDC, 0x60, 0x3A,
    0x60, 0xB8, 0x60, 0xA8, 0x61, 0xBF, 0x61, 0xB6,
    0x61, 0xBF, 0x62, 0x14, 0x62, 0x59, 0x62, 0xB8,
    0x62, 0xA1, 0x63, 0x69, 0x64, 0x3B, 0x66, 0x09,
    0x66, 0x87, 0x67, 0x0B, 0x67, 0x1A, 0x67, 0x29,
    0x68, 0x87, 0x68, 0x99, 0x68, 0xA7, 0x68, 0x82
};

const size_t LutSize = sizeof(LutByteValues);
const unsigned char * pLutBytes = &LutByteValues[0];
```

6.2.3 Required initialization I²C transmission – version p2.22

Initialization version p2.22 is a required part of the V102 device startup.

Initialization version p2.22 offers the following updates over earlier releases:

Performance

- Improved performance at FM 104 MHz for digital radio = off
- Improved performance for FM 96.9 MHz HD radio reception
- Corrected level extension for maximum AM RF AGC step

Initialization version p2.22 includes all the advantages offered from earlier versions:

Fix

- p2.19: Supporting manual selection of the digital radio sample rate ([3.26 FM / AM cmd 84 Set DR Options](#)); not supported for p2.17.
- p2.15: AM Shortwave use: resolved failing reception after switching from FM 94.1 MHz to AM Shortwave band
- p2.15: External FM AGC application and AFU, Jump or Check tuning: resolved wrong mute slope after internal FM AGC step
- p2.14: Radio tuning robustness, resolving issue of wrong quality read and muted audio after > 65000 tunings with less than 32 ms between tuning actions
- p2.14: Erroneous level result in high signal conditions (in certain cases depending on RF AGC threshold, level step correction and level offset settings)
- p2.14: Resolved potential issue of boot state robustness
- p2.13: Tuning action sequence FM Check – FM Preset and FM Check – FM Search; resolved issue of frozen weak signal handling
- p2.13: Tuning action sequence FM Check – FM Jump – ... – FM Check; proper weak signal handling start at closing of FM Check (by Jump, AFU or End action)
- p2.12: Resolved critical issue of muted FM radio after temperature change
- p2.12: Resolved critical issue on variant read
- p2.12: Resolved issue concerning AM noise blanker false triggering on specific condition of 30 kHz adjacent signal with HD radio digital modulation
- p2.12: Improved digital radio large signal handling

New feature

- p2.21: Extension of noise blanker control options ([3.9](#), [3.10](#), [3.11](#), [3.12](#))
- p2.17: Extension of FM automatic bandwidth control options ([3.3 FM / AM cmd 10 Set Bandwidth](#))
- p2.13: Minimum IF bandwidth control ([3.3 FM / AM cmd 10 Set Bandwidth](#))
- p2.09: AM noise detector ([4.1 FM / AM cmd 128 / 129 Get Quality](#))
- p2.09: AM co-channel detector ([3.6 AM cmd 14 Set CoChannelDet](#) and [4.1](#))

Performance

- p2.19: Avoidance of digital radio BCK harmonics in FM digital radio bandwidth.
- p2.17: Improved spurious performance for multiple tuner application, particularly with SAF775X.
- p2.15: FM multipath suppression and AFU, Jump tuning: resolved audio amplitude disturbance after Jump / AFU tuning from high to low signal condition
- p2.13: Optimized digital radio large signal handling timing and adapted threshold
- p2.10: Reduction of potential whistle in the AM SW band at the crystal frequency
- p2.10: Improved digital radio high level signal response and fast settling at tuning
- p2.09: Avoidance of potential whistle at 77.40 MHz

Next to the above published advantages the behavior is otherwise equivalent to the previous device version V101 with p01.24 initialization but including:

- Resolved control issue regarding disabling of FM Highcut 'deemphasis' option

6.3 Operation state description

6.3.1 Boot state

After power-on the device is found in 'boot state', controlled by the integrated power-on-reset detection.

Parallel to the supply power detection a clock signal detector is present which ensures boot state is entered only when a clock signal is present in order to safeguard the external clock application case.

In case of a harmful disruption of the power supply voltage (power dip) or reference clock signal the present operation state will be reset and boot state will be available after power supply voltage and reference signal are re-established.

In boot state the device functionality is reduced to the bare minimum defined by hardware only.

Required additions to the built-in firmware can be loaded up from the host μ C to the internal RAM of the device.

'Boot state' is identified by

- The I²C bus is active; I²C transmissions are acknowledged by the device.
(Note: the I²C device address has been determined during power-on).
- The device is in full power-down.
- GPIO and digital output pins are 'open'.
- No API commands are available; only hardware support.
API commands are ignored and do not harm the boot state operation.
- Reading APPL_Get_Operation_Status returns 'boot state' (0).
Note: This API read operation is defined by the fact that any API read will return '0' here.
This property can be used by the radio system μ C to detect an unforeseen power dip occurrence or unforeseen disruption of the (external) clock reference signal.
- The device can receive certain dedicated hardware commands and data;
Device version V102 requires an initialization at this point for function and performance.
- Command 'Start' will bring the device into 'idle state'.
Note: The 'Start' command is made up of I²C data byte values "14h 00h 01h".
In API definition style this equals a transmission of module = 20, cmd = 0, index = 1 and no parameters.

6.3.2 Idle state

Changing 'boot state' to 'idle state' takes a time of less than 50 ms.

Presence of 'idle state' can be verified by APPL_Get_Operation_Status returning 'idle state' (status = 1).

In 'idle state' very limited control functionality is available, only specific API commands required for system definition are available here.

'Idle state' is identified by

- The I²C bus is active; I²C transmissions are acknowledged by the device.